Frequency-Inspired Contrastive Feature Perturbation for Remote Sensing Semantic Segmentation

Dongsheng Li, *Student Member, IEEE*, Pingting Hao, Xinyuan Guo, *Student Member, IEEE*, Huijie Zhang, Yiming Lin, and Tong Gao, *Member, IEEE*

***Abstract*—Semantic segmentation models for remote sensing images (RSIs) trained under the pretraining–fine-tuning paradigm commonly suffer from limited generalization. From a frequency-domain perspective, we identify the inherent low-frequency bias in deep neural networks as an important contributing factor. Such bias drives pretrained models to overrely on low-frequency appearance statistics and weakens the transfer of high-frequency structural cues. Through spectral response analysis across representative pretraining paradigms, we observe that the segment anything model (SAM)'s prompt-driven pretraining better preserves high-frequency structural cues, offering a favorable initialization; however, low-frequency bias may still be reinforced during fine-tuning as the model fits domain-specific style statistics. To address this issue, we propose frequency-inspired contrastive feature perturbation for SAM (CFP-SAM). This frequency-aware fine-tuning framework introduces a learnable low-frequency perturbation (LLFP) module into SAM's encoder feature space. CFP-SAM jointly maximizes style discrepancy and enforces semantic consistency to suppress low-frequency style dependence and promote transferable structural representations. Extensive experiments on four public benchmarks show that CFP-SAM improves in-domain segmentation accuracy and source-only cross-domain generalization without introducing additional inference overhead. The code is available at https://github.com/wudi-ldd/CFP-SAM**

***Index Terms*—Domain generalization, feature perturbation, parameter-efficient fine-tuning (PEFT), remote sensing, segment anything model (SAM), semantic segmentation.**

I. Introduction

SEMANTIC segmentation of remote sensing images (RSIs) has achieved substantial progress with the development of deep learning, and many recent methods commonly adopt the pretraining–fine-tuning paradigm [1], [2], [3]. In this paradigm, a backbone network pretrained on large-scale natural-image datasets, such as ImageNet-1K [4], is used as an initialization to transfer general visual representations to downstream remote sensing tasks, thereby reducing reliance on costly pixel-level annotations.

However, transferring natural-image-pretrained representations to remote sensing segmentation faces a critical mismatch in feature dependence patterns [3]. Models pretrained with classification or reconstruction objectives tend to rely on easily optimized global appearance cues, such as holistic color patterns and coarse spatial layouts [5]. On the contrary, remote sensing segmentation relies more heavily on fine-grained structural information, including precise boundaries and local textures. As illustrated in Fig. 1, across ViT-L-scale backbones, different pretraining paradigms significantly shape the structural sensitivity of the learned representations. Classification [6] and masked autoencoder (MAE) [7] objectives yield diffuse, globally dominated activations. By comparison, the segment anything model (SAM) [8], [9], pretrained for prompt-driven interactive segmentation, produces boundary-concentrated responses that more closely align with the structural demands of remote sensing segmentation.

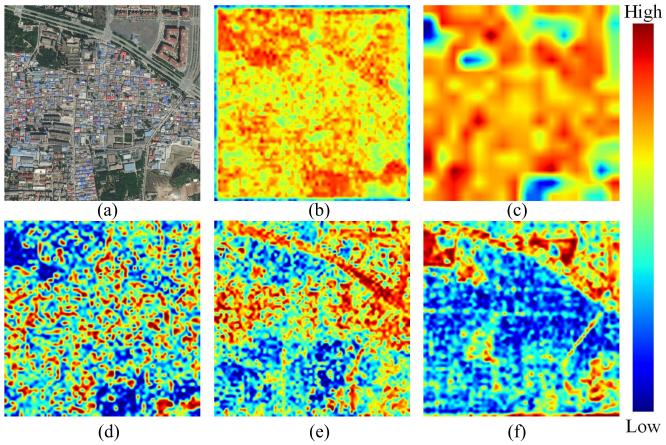


Fig. 1. Feature activation patterns of different ViT-L pretrained backbones on the same RSI. Classification and MAE mainly emphasize coarse regional appearance, whereas SAMRS and DINO produce stronger but more fragmented local responses. SAM exhibits clearer boundary- and layout-aware activations, suggesting a more suitable structural initialization for remote sensing semantic segmentation. Warmer colors indicate higher response intensity. (a) RSI. (b) Classification. (c) Masked autoencoder. (d) SAMRS. (e) DINO. (f) SAM.

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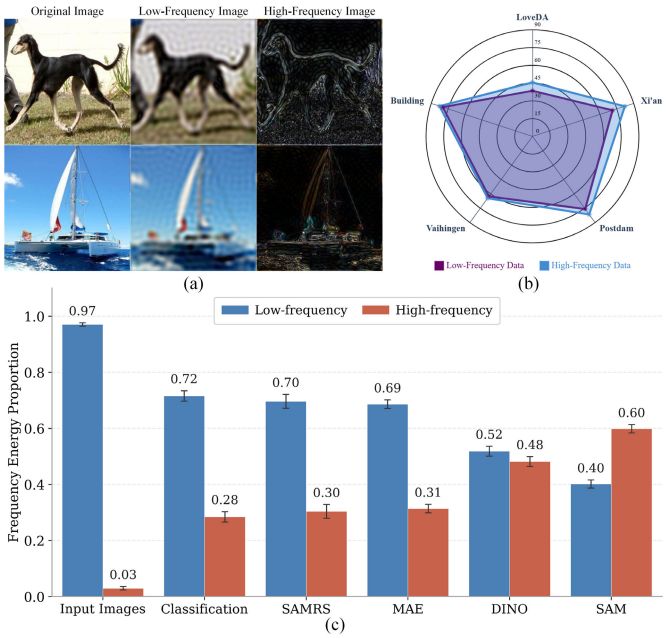


Fig. 2. Frequency-domain analysis of pretraining transferability. (a) Mini-ImageNet images and their low- and high-frequency decompositions. (b) Transfer mIoU of the same ResNet-50 architecture pretrained on low- and high-frequency Mini-ImageNet subsets, showing stronger downstream transfer from high-frequency pretraining. (c) Frequency energy proportions of encoder features on the Vaihingen dataset under different pretraining paradigms, where SAM exhibits the highest high-frequency energy proportion.

This observation motivates us to examine, from a frequency-domain perspective, which pretraining properties are more conducive to transfer for remote sensing segmentation.

To investigate this mismatch from a mechanistic perspective, we turn to frequency-domain analysis [10]. Deep neural networks are known to exhibit spectral bias, tending to fit low-frequency components first during training [11], [12]. This bias can encourage models to rely on easily optimized global appearance cues rather than high-frequency structural details required by remote sensing segmentation. To empirically examine this hypothesis, we pretrain the same ResNet-50 architecture on high-frequency-dominant and low-frequency-dominant subsets of Mini-ImageNet [13], and evaluate their transfer performance on downstream remote sensing segmentation benchmarks [14], [15], [16]. As shown in Fig. 2(a) and (b), high-frequency pretraining consistently yields stronger transfer, indicating that high-frequency structural cues are more transferable to remote sensing tasks than low-frequency appearance statistics, which tend to encode domain-specific factors that vary across sensors and regions [17]. Fig. 2(c) further shows the comparison of the frequency energy proportions of encoder features across different pretraining paradigms on the Vaihingen dataset [18]. Features from classification, MAE, and SAMRS-based segmentation pretraining [19] are dominated by low-frequency energy. On the contrary, SAM exhibits the highest proportion of high-frequency energy, suggesting that its pretraining objective more naturally aligns with the structural demands of remote sensing segmentation.

Unlike models pretrained with classification or reconstruction objectives, SAM is pretrained for prompt-driven interactive segmentation. The prompt provides an explicit spatial prior for the target region, reducing the need to rely on low-frequency appearance cues for coarse localization. As a result, SAM places greater emphasis on fine-grained structural evidence, including object boundaries and local textures, making it a more suitable initialization for remote sensing segmentation.

Although SAM provides a structurally aware initialization, low-frequency style dependence can still be reinforced during fine-tuning, where the model tends to overfit domain-specific appearance statistics that do not generalize across scenes. To address this, we propose frequency-inspired contrastive feature perturbation for SAM (CFP-SAM). This frequency-aware fine-tuning framework introduces a learnable low-frequency perturbation (LLFP) module into the encoder's feature space during training, generating diverse low-frequency style variations to reduce dependence on style shortcuts. We further maximize the style discrepancy between clean and perturbed branches while enforcing semantic consistency, so that low-frequency perturbations serve as robustness training signals without inducing semantic drift. The main contributions of this article are as follows.

1) We provide a frequency-domain analysis of pretraining transferability for remote sensing segmentation, indicating that high-frequency structural cues are more transferable and that SAM provides a favorable initialization for remote sensing dense prediction.
2) We propose CFP-SAM, a frequency-aware fine-tuning framework built on SAM, which introduces LLFPs and dual constraints to suppress style shortcuts and enhance transferable representation learning.
3) Extensive experiments on four public benchmarks demonstrate that CFP-SAM improves both in-domain segmentation and source-only cross-domain generalization without introducing additional inference overhead.

II. Related Work

A. RSI Semantic Segmentation

Semantic segmentation of RSIs is a core task in remote sensing scene understanding. Early methods largely followed natural-image segmentation frameworks, with representative architectures including fully convolutional networks (FCNs) [20] and the DeepLab series [21]. These methods laid the foundation for pixel-level dense prediction through fully convolutional or encoder–decoder architectures. With the widespread adoption of Transformers for visual tasks, architectures such as SegFormer [22] further advanced global context modeling for semantic segmentation and were subsequently adapted to remote sensing segmentation.

RSIs present distinctive challenges, including large variations in object scale, complex object boundaries, and interactions between local textures and global context. To address these challenges, researchers have developed a range of segmentation networks tailored for remote sensing. Representative methods include UNetFormer [23], MANET [24],

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A^2-feature pyramid network (FPN) [25], and CMTFNet [26], which mainly focus on multiscale perception, local–global interaction, and fine-grained feature modeling. More recent work has further advanced remote sensing segmentation from complementary perspectives. CenterSeg [27] introduced a prototype-based classifier with Grassmann manifold regularization to mitigate intraclass variance. SparseFormer [28] addressed point-supervised segmentation by leveraging credible pseudolabels and knowledge distillation. SCSM [29] reconstructed spatial attention through scene coupling and local–global semantic masks to model geospatial object correlations. These segmentation methods, as well as related remote sensing studies on image reconstruction [30], [31], [32], mainly advance task-specific representation learning. On the contrary, the transfer behavior of pretrained representations and their adaptation bias remain less explored.

B. Visual Foundation Models and PEFT

Large-scale visual foundation models provide reusable visual priors for dense prediction and have recently been explored for remote sensing segmentation. SAM [8], [9] learned general segmentation priors through large-scale interactive pretraining and has been adapted to various remote sensing tasks. RSPrompter [33] transferred SAM to remote sensing instance segmentation through learned prompt generation. SAM-Assisted [34] incorporated SAM-generated object regions and boundary information as auxiliary supervision for semantic segmentation. MSA-SAM [35] integrated pyramid and window multiscale adapters into the SAM image encoder to improve the perception of scale-varying targets, and unified multimodal fine-tuning further explored adapter- and low-rank adaptation (LoRA)-based SAM adaptation for remote sensing semantic segmentation [36]. UV-SAM [15] used a specialized segmentation model to generate mixed prompts for SAM to identify urban-village boundaries. SAMRS [19] leveraged SAM and existing remote sensing detection datasets to construct a large-scale segmentation dataset and demonstrated the value of segmentation pretraining for downstream adaptation.

In parallel, remote sensing foundation models have been developed to improve representation learning through remote sensing-domain or task-aligned pretraining. Representative examples include RingMo [3], SatMAE [37], Scale-MAE [38], and MTP [39]. To reduce the cost of adapting foundation models, parameter-efficient fine-tuning (PEFT) methods, such as adapters [40] and LoRA [41], replace full parameter updates with lightweight task-specific adjustments and have been widely used for adapting visual foundation models to downstream segmentation tasks [35]. However, PEFT usually operates on frozen or largely frozen pretrained backbones, and the biases inherited from pretraining can limit its effectiveness. When a pretrained model retains low-frequency style dependence, limited trainable updates may leave this bias insufficiently corrected and even reinforce it during downstream adaptation. How to suppress low-frequency style dependence under parameter-efficient SAM adaptation remains underexplored.

C. Frequency-Domain Learning and Domain Generalization

Frequency-domain analysis provides an important perspective for understanding the learning biases of deep neural networks. The frequency principle [13], [17] suggests that neural networks tend to fit low-frequency components earlier during training, reflecting an inherent low-frequency bias. Related studies further show that this bias may encourage models to rely on low-frequency appearance statistics as shortcut cues [10], [12], [42], weakening their ability to capture high-frequency boundary and texture cues. This issue is particularly relevant to remote sensing segmentation, where dense prediction requires both stable contextual understanding and fine-grained structural delineation.

Frequency-aware modeling has therefore been increasingly explored in remote sensing semantic segmentation. DualM [43] introduced a spatial–frequency dual-domain Transformer to improve boundary modeling for urban RSIs. SFFNet [44] used wavelet-based spatial–frequency fusion to combine spatial details with frequency-domain representations. AFENet [45] adaptively separated and modulated high- and low-frequency components according to image content. SynFSNet [46], built upon AFSIFormer blocks, embedded frequency information into the attention mechanism through adaptive frequency–spatial interaction and residual coupling. These methods demonstrate that frequency-domain cues are effective for enhancing feature representation and boundary perception in remote sensing segmentation. However, their frequency operations mainly serve to enhance forward features rather than suppress style dependence during fine-tuning.

Perturbation-based domain generalization methods are also related to our work. AdaIN [47] represented visual style using channelwise feature statistics, while MixStyle [48] and DSU [49] expanded the training style space by mixing or sampling feature statistics. CCFP [50] learned latent feature perturbations with semantic-consistency constraints. Beyond feature-statistics perturbation, Jeon et al. [51] decomposed features into high- and low-frequency components and stylized the low-frequency part while preserving high-frequency structure. Liu et al. [52] proposed a learnable frequency-domain perturbation strategy for single-source domain generalization, and CrossEarth [53] introduced earth-style injection to improve cross-region generalization in remote sensing segmentation. On the contrary, CFP-SAM applies learnable perturbations to the low-frequency amplitudes of SAM encoder features as a training-time regularization signal. This design suppresses low-frequency style dependence while preserving SAM's structural priors and introduces no additional frequency modules into the inference path.

III. Method

A. Problem Formulation and Method Overview

1) Task Definition: Let $\mathcal{D}_s = \{(\mathbf{x}_i, \mathbf{y}_i)\}_{i=1}^{N}$ denote a labeled source-domain remote sensing semantic segmentation dataset, where $\mathbf{x}_i \in \mathbb{R}^{H\times W\times 3}$ is the input image and $\mathbf{y}_i \in \{1,\ldots,K\}^{H\times W}$ is the pixelwise semantic annotation with K categories. Unseen target-domain datasets contain only test images and are not

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used during training. Under this source-only adaptation setting, we aim to learn a semantic segmentation model $f_\theta : \mathbb{R}^{H\times W\times 3} \to \mathbb{R}^{H\times W\times K}$ that produces per-pixel category logits. The goal is to achieve accurate segmentation on the source domain while maintaining strong generalization to unseen target domains.

2) Core Challenge: Many existing methods adopt the pretraining–fine-tuning paradigm, where the encoder is initialized with parameters θ_{pre} pretrained on large-scale natural-image datasets and then fine-tuned on the source-domain dataset $\mathcal{D}_s$. However, as discussed in Section I, natural images and RSIs exhibit significant differences in visual distributions and spatial structures. More critically, the inherent low-frequency bias of deep neural networks makes models prone to fitting easily optimized low-frequency appearance features rather than the structural cues required for remote sensing segmentation. This bias can be further reinforced during fine-tuning, leading to over-reliance on domain-specific low-frequency style patterns and limiting source-only cross-domain generalization.

3) Method Overview: To address the above challenges, we propose CFP-SAM, a frequency-aware fine-tuning framework for SAM, as illustrated in Fig. 3. The framework consists of three key components.

1) *SAM-based encoder initialization*, which provides a structurally sensitive starting point with stronger high-frequency responses.
2) LLFP module, which dynamically perturbs low-frequency components in the frequency domain of encoder features to generate diverse low-frequency style variations.
3) *Dual-constraint optimization*, which maximizes the Gram-matrix-based style discrepancy between clean and perturbed features to explore challenging low-frequency style variations while enforcing semantic consistency between the two branches to preserve stable predictions under perturbation.

This design reduces the model's reliance on low-frequency style patterns and encourages transferable structural representations. The perturbation branch is disabled at inference, so the training-time LLFP mechanism introduces no additional inference overhead.

B. SAM-Based Encoder Initialization

As shown by the frequency-domain analysis in Section I, pretraining under the SAM segmentation paradigm preserves stronger high-frequency structural responses than classification- or reconstruction-based pretraining. Motivated by this observation, we adopt a Hiera-Large image encoder pretrained under the SAM segmentation paradigm as the backbone encoder. However, SAM's original mask decoder produces prompt-conditioned, category-agnostic masks and cannot be directly applied to dense multiclass semantic segmentation in remote sensing scenarios. Therefore, we retain the pretrained image encoder and replace the original prompt-driven decoder with a lightweight semantic segmentation head.

Lightweight Segmentation Head Adaptation: To adapt SAM to remote sensing semantic segmentation, we replace its prompt-conditioned mask decoder with a lightweight segmentation head for prompt-free dense multiclass prediction. As illustrated in Fig. 4, the adopted Hiera-Large encoder produces multiscale stage features $\{f_1, f_2, f_3, f_4\}$ at 1/4, 1/8, 1/16, and 1/32 resolutions, with channel dimensions of 144, 288, 576, and 1152, respectively. These features are fed into a FPN neck, which projects them into a unified 256-channel feature pyramid for multiscale fusion. The lower resolution features are upsampled to a common 1/4 resolution, smoothed by 3×3 convolutions, and aggregated by elementwise summation to preserve complementary spatial details across scales. The fused representation is finally decoded into K-class semantic logits through a compact convolutional predictor. This design enables prompt-free dense prediction while avoiding the task-misaligned prompt-based decoding of SAM's original decoder.

Knowledge-Preserving Adaptation via LoRA: While SAM provides high-quality initialization, its large parameter count makes full fine-tuning computationally expensive. More importantly, standard fine-tuning is prone to overfitting source-domain appearance statistics, potentially weakening the transferable structural priors established during pretraining. To address this, we freeze the SAM encoder weights and introduce LoRA [41] into the attention layers to enable parameter-efficient adaptation with minimal trainable parameters.

In particular, for query and value projection matrices $\mathbf{W}_q, \mathbf{W}_v \in \mathbb{R}^{d\times d}$ in each attention module, we introduce trainable low-rank updates

$$\mathbf{W}'_q = \mathbf{W}_q + \mathbf{B}_q\mathbf{A}_q, \quad \mathbf{W}'_v = \mathbf{W}_v + \mathbf{B}_v\mathbf{A}_v \tag{1}$$

where $\mathbf{A}_q, \mathbf{A}_v \in \mathbb{R}^{r\times d}$ and $\mathbf{B}_q, \mathbf{B}_v \in \mathbb{R}^{d\times r}$ are trainable low-rank matrices with rank $r \ll d$. This design preserves SAM's pretrained representations while enabling lightweight adaptation to remote sensing distributions through low-rank increments. During inference, the LoRA weights can be merged into the original projection matrices, introducing no additional inference overhead.

C. LLFP Module

LLFP builds on SAM's preserved high-frequency structural priors without directly amplifying them. By perturbing low-frequency amplitudes during training, LLFP renders low-frequency style statistics unreliable; the semantic-consistency constraint then encourages stable predictions that rely more on high-frequency structural representations.

Frequency-Domain Decomposition and Low-Frequency Identification: Given intermediate feature maps $\mathbf{X} \in \mathbb{R}^{B\times C\times H\times W}$ from the image encoder, we apply a 2-D Fourier transform over the spatial dimensions and shift the zero-frequency component to the center

$$\mathbf{Z} = \text{fftshift}\left(\mathcal{F}(\mathbf{X})\right), \quad \mathbf{A} = |\mathbf{Z}|, \quad \mathbf{P} = \angle\mathbf{Z} \tag{2}$$

where $\mathbf{Z} \in \mathbb{C}^{B\times C\times H\times W}$ is the centered complex spectrum, and $\mathbf{A}$ and $\mathbf{P} \in \mathbb{R}^{B\times C\times H\times W}$ denote the amplitude and phase spectra, respectively. As discussed in Section I, low-frequency

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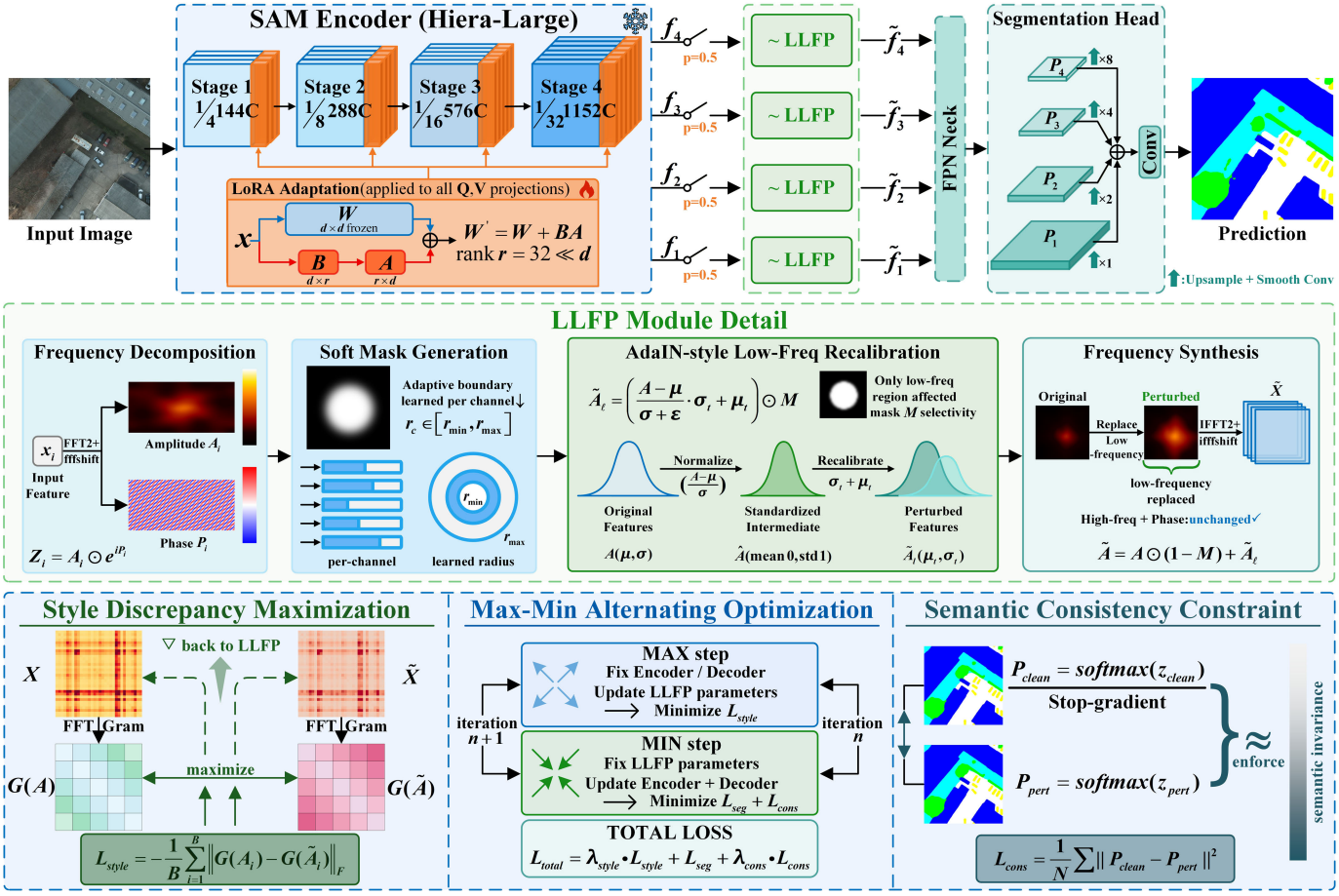


Fig. 3. Overview of CFP-SAM. LLFP perturbs multiscale encoder features during training, while the clean and perturbed branches are optimized through style-discrepancy maximization and semantic consistency. LLFP is disabled at inference.

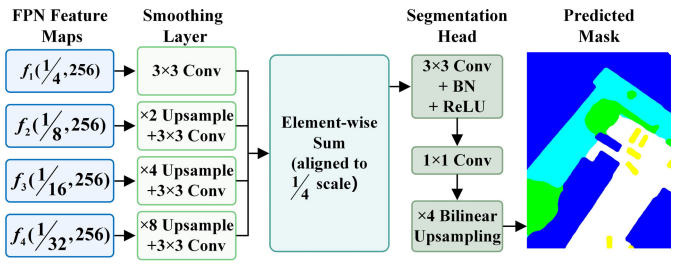


Fig. 4. Architecture of the lightweight multiscale segmentation head.

amplitude components tend to encode global appearance and style information, whereas high-frequency components capture structural cues such as boundaries and textures. Therefore, LLFP perturbs only low-frequency amplitudes while preserving high-frequency components and the phase spectrum, thereby maintaining structural cues.

To achieve adaptive separation between low- and high-frequency components, we introduce learnable unconstrained parameters $\{\rho_c\}_{c=1}^{C}$ and map them to per-channel low-frequency radii

$$r_c = r_{\min} + (r_{\max} - r_{\min})\,\sigma(\rho_c) \tag{3}$$

where $\sigma(\cdot)$ denotes the sigmoid function, so that $r_c \in [r_{\min}, r_{\max}]$. We convert the proportional radii to pixel-scale radii as $r_c^{\text{pix}} = r_c \cdot \min(H, W)$. Based on these radii, we construct a raised-cosine soft mask $\mathbf{M} \in \mathbb{R}^{1\times C\times H\times W}$ to smoothly select low-frequency regions, which is broadcast across the batch dimension. Let $t_c(h,w) = \text{clip}((d(h,w) - r_c^{\text{pix}})/w_c, 0, 1)$ denote the normalized transition distance. The mask is defined elementwise as

$$[\mathbf{M}]_{1,c,h,w} = \frac{1}{2}\left(1 + \cos\left(\pi\, t_c\,(h,w)\right)\right) \tag{4}$$

where $d(h,w)$ denotes the Euclidean distance from frequency coordinate (h,w) to the spectrum center and $w_c = \max(\alpha r_c^{\text{pix}}, w_{\min})$ is the transition bandwidth.

LLFP: Motivated by the observation that feature statistics are closely associated with style variations [47], LLFP introduces learnable modulation parameters to synthesize perturbed style statistics from source-domain encoder features without requiring target-domain samples. In particular, we first compute weighted statistics within the low-frequency region

$$\mu = \frac{\sum_{h,w} \mathbf{A} \odot \mathbf{M}}{\sum_{h,w} \mathbf{M} + \varepsilon}$$
$$\sigma = \sqrt{\frac{\sum_{h,w} (\mathbf{A} - \mu)^2 \odot \mathbf{M}}{\sum_{h,w} \mathbf{M} + \varepsilon} + \varepsilon} \tag{5}$$

where $\odot$ denotes elementwise multiplication with broadcasting and $\mu, \sigma \in \mathbb{R}^{B\times C\times 1\times 1}$ denotes the low-frequency

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amplitude statistics. We introduce learnable per-channel parameters $\hat{\gamma}, \hat{\beta} \in \mathbb{R}^C$ and bound them through tanh reparameterization to stabilize optimization

$$\gamma = \alpha_\gamma \tanh(\hat{\gamma}), \qquad \beta = \alpha_\beta \tanh\left(\hat{\beta}\right) \tag{6}$$

where $\alpha_\gamma, \alpha_\beta > 0$ is fixed scalar hyperparameters controlling the maximum perturbation magnitude. The perturbed statistics are defined as

$$\mu_p = \mu + \beta \odot \sigma, \qquad \sigma_p = \sigma \odot \exp(\gamma) \tag{7}$$

where the multiplicative form of σ_p guarantees positivity and confines the scale ratio $\sigma_p/\sigma = \exp(\gamma)$ to $(\exp(-\alpha_\gamma), \exp(\alpha_\gamma))$, while the additive shift $\mu_p - \mu = \beta\sigma$ is bounded within $\pm\alpha_\beta\sigma$ during gradient-ascent optimization. At initialization ($\hat{\gamma} = \hat{\beta} = \mathbf{0}$), we have $\mu_p = \mu$ and $\sigma_p = \sigma$, recovering an identity mapping that provides a stable starting point for perturbation learning.

The low-frequency amplitudes are then recalibrated by

$$\tilde{\mathbf{A}}_\ell = \mathrm{ReLU}\left(\frac{\mathbf{A} - \mu}{\sigma + \varepsilon} \odot \sigma_p + \mu_p\right) \odot \mathbf{M} \tag{8}$$

where the ReLU operator ensures that the recalibrated amplitudes remain nonnegative. We then replace the low-frequency amplitudes while preserving high-frequency components and the original phase

$$\tilde{\mathbf{A}} = \mathbf{A} \odot (1 - \mathbf{M}) + \tilde{\mathbf{A}}_\ell. \tag{9}$$

The perturbed features are then reconstructed via inverse Fourier transform

$$\tilde{\mathbf{Z}} = \tilde{\mathbf{A}} \odot \exp(j\mathbf{P}), \qquad \tilde{\mathbf{X}} = \Re\left(\mathcal{F}^{-1}\left(\mathrm{ifftshift}\left(\tilde{\mathbf{Z}}\right)\right)\right) \tag{10}$$

where j denotes the imaginary unit.

LLFP is applied to the multiscale encoder features $\{f_1, f_2, f_3, f_4\}$ with scale-specific learnable parameters to generate diverse low-frequency style perturbations. During training, each LLFP module is independently activated with probability $p = 0.5$, yielding a clean branch with LLFP bypassed and a perturbed branch with LLFP applied to the sampled scales. Since independent sampling may deactivate all scales, we use fallback activation before optimization by randomly forcing one LLFP scale to be active. This prevents the perturbed branch from degenerating into the clean branch and provides a valid perturbed view for both discrepancy maximization and semantic regularization. During inference, LLFP is disabled, and the perturbed branch is discarded, introducing no additional inference cost.

D. Dual-Constraint Optimization

LLFP generates style-perturbed features by modulating low-frequency amplitudes. To make these perturbations beneficial for robust training, we introduce a dual-constraint optimization strategy. LLFP is optimized to enlarge the style discrepancy between the clean and perturbed branches, while the segmentation model is trained to maintain consistent semantic predictions under these perturbations.

Gram-Based Style Discrepancy: Inspired by style representation learning, we use Gram matrices to measure second-order channel correlations between clean and perturbed features, characterizing global appearance statistics while discarding spatial correspondence. Since LLFP operates on low-frequency amplitudes, the discrepancy is computed on masked amplitude spectra. Because the per-channel low-frequency radius affects the effective window size, a standard Gram matrix would couple the magnitude of the discrepancy to the number of selected frequency components. We therefore apply unit-area-normalized weighting to remove this size dependence and make the discrepancy comparable across channels and scales.

For any amplitude tensor $\mathbf{U} \in \{\mathbf{A}, \tilde{\mathbf{A}}\}$, its masked and normalized representation is defined as

$$\mathbf{U}_w = \mathbf{U} \odot \frac{\sqrt{\mathbf{M}}}{\sqrt{\mathbf{S} + \varepsilon}}, \qquad \mathbf{S} = \frac{1}{HW}\sum_{h,w}\mathbf{M} \tag{11}$$

where $\mathbf{S} \in \mathbb{R}^{1\times C\times 1\times 1}$ denotes the area fraction of the low-frequency mask and is broadcast to the batch and spatial dimensions. After reshaping $\mathbf{U}_w$ to $\mathbb{R}^{B\times C\times HW}$, the Gram matrix for the ith sample is computed as

$$\mathbf{G}(\mathbf{U}_i) = \frac{1}{CHW}\mathbf{U}_{w,i}\mathbf{U}_{w,i}^{\top} \in \mathbb{R}^{C\times C}. \tag{12}$$

The style discrepancy between the clean and perturbed branches is defined as

$$\mathcal{D}_{\mathrm{style}} = \frac{1}{B}\sum_{i=1}^{B}\left\|\mathbf{G}(\mathbf{A}_i) - \mathbf{G}\left(\tilde{\mathbf{A}}_i\right)\right\|_F. \tag{13}$$

A larger $\mathcal{D}_{\mathrm{style}}$ indicates stronger low-frequency style discrepancy between the two branches. For multiscale encoder features, this discrepancy is computed at each selected scale and averaged across them.

Semantic-Consistency Constraint: To prevent style perturbations from altering semantic predictions, we impose a consistency constraint between the clean and perturbed branches. Let $\mathbf{z}$ and $\tilde{\mathbf{z}}$ denote the segmentation logits from the clean and perturbed branches, respectively. The clean prediction is treated as a teacher by stopping its gradient, and the semantic-consistency loss is defined as

$$\mathcal{L}_{\mathrm{cons}} = \frac{1}{BHW}\left\|\mathrm{sg}(\mathrm{softmax}(\mathbf{z})) - \mathrm{softmax}(\tilde{\mathbf{z}})\right\|_F^2 \tag{14}$$

where softmax$(\cdot)$ is applied along the class dimension and sg$(\cdot)$ denotes the stop-gradient operation.

Overall Objective and Optimization: Segmentation supervision is applied to both branches

$$\mathcal{L}_{\mathrm{seg}} = \mathcal{L}_{\mathrm{sup}}(\mathbf{z}, \mathbf{y}) + \mathcal{L}_{\mathrm{sup}}(\tilde{\mathbf{z}}, \mathbf{y}), \qquad \mathcal{L}_{\mathrm{sup}} = \mathcal{L}_{\mathrm{ce}} + \mathcal{L}_{\mathrm{dice}}. \tag{15}$$

Let ϕ denote the parameters of LLFP, and let θ denote the trainable parameters of LoRA and the segmentation head. Training proceeds through the following alternating objectives:

$$\begin{aligned} \max_{\phi} \quad & \lambda_{\mathrm{style}}\mathcal{D}_{\mathrm{style}} \\ \min_{\theta} \quad & \mathcal{L}_{\mathrm{seg}} + \lambda_{\mathrm{cons}}\mathcal{L}_{\mathrm{cons}}. \end{aligned} \tag{16}$$

With θ fixed, we first update ϕ by gradient ascent to maximize the style discrepancy and search for challenging low-frequency perturbations. If stochastic LLFP activation produces no effective perturbation, fallback activation is

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Algorithm 1 CFP-SAM Training Iteration

Require: Mini-batch $(\mathbf{x}, \mathbf{y})$; parameters θ, ϕ; weights λ_{style}, λ_{cons}
Ensure: Updated ϕ, updated θ
Step 1: maximize style discrepancy (update ϕ)
Freeze θ (`requires_grad=False`); enable gradients for ϕ
Compute clean features under `torch.no_grad()`; compute perturbed features with frozen θ
$\mathcal{L}_{\max} \leftarrow -\lambda_{\text{style}} \mathcal{D}_{\text{style}}$; update ϕ by descending $\nabla_\phi \mathcal{L}_{\max}$
Unfreeze θ
Step 2: minimize segmentation and consistency losses (update θ)
Freeze ϕ (`requires_grad=False`); enable gradients for θ
Recompute $\mathbf{z}$ and $\tilde{\mathbf{z}}$ through new clean and perturbed forward passes
$\mathcal{L}_{\min} \leftarrow \mathcal{L}_{\text{sup}}(\mathbf{z}, \mathbf{y}) + \mathcal{L}_{\text{sup}}(\tilde{\mathbf{z}}, \mathbf{y}) + \lambda_{\text{cons}} \mathcal{L}_{\text{cons}}$, with $\mathcal{L}_{\text{cons}}$ defined in Eq. (14); update θ by descending $\nabla_\theta \mathcal{L}_{\min}$
Unfreeze ϕ

applied before computing $\mathcal{D}_{\text{style}}$. With ϕ fixed, we then update θ by gradient descent using the segmentation and consistency losses, encouraging stable semantic predictions under these perturbations. Algorithm 1 summarizes one training iteration, where $\text{sg}(\cdot)$ denotes the stop-gradient operation.

IV. Experiments

To evaluate CFP-SAM, we conduct experiments addressing four research questions.

RQ1: Does SAM's prompt-driven segmentation pretraining provide superior initialization for remote sensing semantic segmentation compared with other pretraining paradigms?

RQ2: How does CFP-SAM compare with state-of-the-art remote sensing semantic segmentation methods?

RQ3: How do the core modules of CFP-SAM affect segmentation performance?

RQ4: To what extent does CFP-SAM improve source-only cross-domain generalization to unseen domains, and how does it affect target-domain feature distributions?

The remainder of this section is organized as follows. We first introduce the experimental setup in Section IV-A, then address RQ1–RQ4 in Sections IV-B–IV-E, respectively.

A. Experimental Setup

Datasets: We evaluate CFP-SAM on four public benchmarks. *ISPRS Vaihingen* [18] consists of 33 high-resolution aerial orthophotos over Vaihingen, Germany, with an average size of 2494 × 2064 pixels and a ground sampling distance (GSD) of 9 cm. The images contain three spectral bands, namely near-infrared, red, and green (IRRG). The annotations include five foreground land-cover categories, namely impervious surfaces, buildings, low vegetation, trees, and cars, together with clutter/background. Following the commonly used split, 16 images are used for training and 17 for testing. Clutter/background is excluded from evaluation. Images are cropped into 1024×1024 patches. *ISPRS Potsdam* [18] consists of 38 orthophotos over Potsdam, Germany, each with a size of 6000 × 6000 pixels and a GSD of 5 cm. It follows the same category definitions as Vaihingen. Only RGB bands are used as input. Following the commonly used split, 23 images are used for training and 15 for testing. Clutter/background is excluded from evaluation. Images are cropped into 1024 × 1024 patches. *Beijing and Xi'an* [15] are binary urban-village segmentation benchmarks covering high-resolution satellite imagery at approximately 1.05 m/pixel. The Beijing subset contains 531 images, and the Xi'an subset contains 205 images. All images have a size of 1024×1024 pixels. The data are split into training, validation, and test sets at a 6:2:2 ratio.

Implementation Details: All experiments are implemented in PyTorch and conducted on a single NVIDIA RTX 4090 GPU. The encoder is initialized with SAM-pretrained Hiera-Large weights and adapted via LoRA with rank $r = 32$ and dropout 0.1. The pretrained encoder is frozen except for the LoRA parameters, while the segmentation head and LLFP parameters are trained from scratch. We use the AdamW optimizer [54] with a learning rate of 1×10^{-4} and weight decay of 1×10^{-2}. The learning rate is scheduled by a three-epoch linear warm-up followed by cosine decay to 1% of the initial learning rate [55]. All models are trained for 100 epochs with a batch size of 2. The loss weights are set to $\lambda_{\text{style}} = \lambda_{\text{cons}} = 1.0$.

For LLFP, the learnable low-frequency radius is constrained to $[r_{\min}, r_{\max}] = [0.05, 0.45]$. The raised-cosine transition bandwidth coefficient is set to $\alpha = 0.25$, with a minimum bandwidth of $w_{\min} = 4.0$ pixels. During training, LLFP is independently activated at each selected encoder scale with probability $p = 0.5$, and fallback activation is used when all scales are inactive. This stochastic activation avoids deterministic multiscale perturbation and increases style diversity across iterations. The modulation parameters $\hat{\gamma}$ and $\hat{\beta}$ are initialized to zero and bounded by the scaled tanh reparameterization, with $\alpha_\gamma = \alpha_\beta = 1.0$. Data augmentation consists of random horizontal and vertical flips applied uniformly across all datasets.

Evaluation Metrics: For ISPRS Vaihingen and Potsdam, we report per-class $F1$-score, mean $F1$-score (mF1), mean intersection over union (mIoU), and overall accuracy (OA). For class c

$$\text{F1}_c = \frac{2\text{TP}_c}{2\text{TP}_c + \text{FP}_c + \text{FN}_c}, \quad \text{IoU}_c = \frac{\text{TP}_c}{\text{TP}_c + \text{FP}_c + \text{FN}_c}$$

mF1 and mIoU are the classwise averages of $F1_c$ and IoU_c, respectively, while OA is the proportion of correctly classified pixels. Clutter/background pixels are excluded from evaluation.

For the Beijing and Xi'an urban-village datasets, we report foreground IoU, $F1$-score, recall, and precision

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}}, \quad \text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}}$$

where TP, FP, and FN are computed for the foreground urban-village class.

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TABLE I

TRANSFERABILITY OF PRETRAINED BACKBONES UNDER FULL FINE-TUNING AND RANK-1 LORA ADAPTATION. BEST RESULTS UNDER FULL FINE-TUNING AND LORA ARE SHOWN IN BOLD AND UNDERLINED, RESPECTIVELY. → +SAMRS DENOTES ADDITIONAL SEGMENTATION PRETRAINING ON SAMRS INITIALIZED FROM SUPERVISED-CLASSIFICATION WEIGHTS

Pretraining	Backbone	Pretraining Data	Scale (Imgs)	FT	Trainable Params (M)	ISPRS				Urban Village			
						Vaihingen		Potsdam		Beijing		Xi'an	
						mIoU	mF1	mIoU	mF1	IoU	F1	IoU	F1
Sup. Cls. [4]	ViT-L/16	ImageNet-1K	1.3M	Full	305.66	80.11	88.70	86.41	92.56	70.88	82.96	76.05	86.40
	ViT-L/16	ImageNet-1K	1.3M	LoRA	2.51	76.50	86.30	84.49	91.44	66.49	79.87	71.10	83.11
	ViT-B/16	ImageNet-1K	1.3M	Full	90.57	77.34	86.89	86.60	92.68	72.26	83.90	75.62	86.12
	ViT-B/16	ImageNet-1K	1.3M	LoRA	1.83	73.16	83.97	81.98	89.95	67.58	80.65	69.62	82.09
	ResNet-50	ImageNet-1K	1.3M	Full	28.23	76.43	86.21	85.06	91.81	70.71	82.84	73.12	84.47
	ResNet-101	ImageNet-1K	1.3M	Full	47.22	76.60	86.35	85.90	92.30	70.53	82.72	74.53	85.41
→ +SAMRS [19]	ViT-B/16	ImageNet-1K + SAMRS	1.3M+0.1M	Full	90.26	72.11	83.16	79.39	88.26	64.84	78.67	61.36	76.05
	ViT-B/16	ImageNet-1K + SAMRS	1.3M+0.1M	LoRA	0.65	69.01	74.33	63.15	76.84	57.34	72.89	41.42	58.57
	ResNet-50	ImageNet-1K + SAMRS	1.3M+0.1M	Full	28.23	73.58	84.28	82.88	90.52	70.18	82.48	72.36	83.96
MAE [7]	ViT-L/16	ImageNet-1K	1.3M	Full	305.66	81.95	89.89	87.61	93.25	72.75	84.23	75.94	86.33
	ViT-L/16	ImageNet-1K	1.3M	LoRA	2.51	79.82	88.49	85.99	92.33	71.77	83.56	74.98	85.70
DINO [56]	ViT-L/14	LVD-142M	142M	Full	306.73	82.24	90.06	86.17	92.42	65.70	79.30	73.16	84.50
	ViT-L/14	LVD-142M	142M	LoRA	2.51	82.48	90.23	87.43	93.17	72.15	83.82	75.56	86.08
SAM [8], [9]	ViT-L/16	SA-1B	11M	Full	312.93	**83.39**	**90.79**	88.04	93.51	73.49	84.72	**80.51**	**89.20**
	ViT-L/16	SA-1B	11M	LoRA	0.74	83.01	90.54	88.43	93.73	72.95	84.36	78.77	88.12
	ViT-B/16	SA-1B	11M	Full	90.26	82.60	90.31	87.52	93.20	71.84	83.61	78.84	88.17
	ViT-B/16	SA-1B	11M	LoRA	0.65	82.59	90.30	85.83	92.25	71.51	83.39	77.45	87.30
	Hiera-L	SA-1B + SA-V	22M	Full	213.29	83.33	90.74	**88.28**	**93.64**	**74.17**	**85.17**	77.59	87.38
	Hiera-L	SA-1B + SA-V	22M	LoRA	0.76	82.59	90.30	87.78	93.37	73.94	85.02	77.63	87.41

B. Effect of Pretraining Paradigms on Transferability (RQ1)

To assess pretraining transferability, we compare supervised classification (Sup. Cls.), reconstruction-based self-supervision (MAE), distillation-based self-supervision (DINO), and prompt-driven segmentation (SAM). All backbones use the same lightweight single-scale head operating on the final encoder output. We evaluate full fine-tuning as a high-capacity reference and rank-1 LoRA as a constrained probe of pretrained representations; the main CFP-SAM experiments instead use rank $r = 32$.

Overall Performance: Table I reports results on all four datasets. SAM-pretrained models achieve the strongest overall performance under both adaptation settings, followed by MAE and DINO in most cases, while supervised classification is generally weaker. Although DINO is pretrained on 142 M images, whereas the SAM ViT models use 11 M images, its transfer performance remains lower across all benchmarks under the evaluated configurations, suggesting that pretraining-objective alignment is an important factor in dense prediction transfer.

Transferability and Mechanistic Interpretation: The gap between full fine-tuning and rank-1 LoRA reflects how readily pretrained representations adapt under a limited parameter budget. On Vaihingen, this gap is 0.38 mIoU for SAM ViT-L/16, compared with 2.13 for MAE and 3.61 for supervised classification. Classification pretraining relies on image-level discrimination through the CLS token or global pooling, and thus does not explicitly optimize pixel-wise spatial correspondence. MAE preserves spatial layout through reconstruction but may favor low-frequency appearance recovery over high-frequency boundary cues. On the contrary, SAM's prompt-conditioned mask generation directly encourages boundary localization and region partitioning, thereby better aligning its representations with dense prediction.

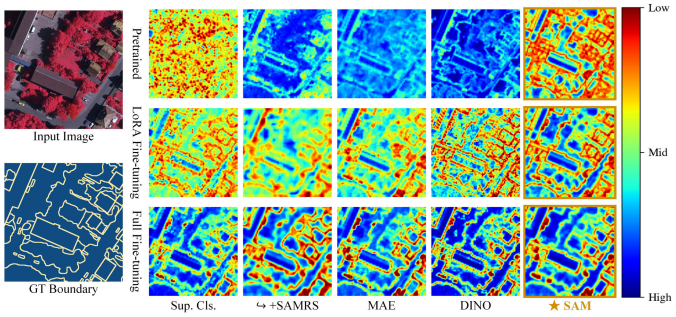


Fig. 5. Feature-gradient energy maps of five pretraining paradigms under three adaptation states on a Vaihingen test image. All maps share the same normalization scale. Warmer colors indicate lower gradient magnitude, while cooler colors indicate higher gradient magnitude. Pretrained, rank-1 LoRA-adapted, and fully fine-tuned results are compared columnwise.

Two-Stage Segmentation Pretraining: Further pretraining on SAMRS does not improve transferability in this setting. Under rank-1 LoRA, ViT-B/16 achieves only 69.01 and 63.15 mIoU on Vaihingen and Potsdam, respectively. A possible explanation is that category-aware SAMRS pretraining strengthens encoder–decoder coupling, which cannot be fully recovered after replacing the original decoder under a constrained LoRA budget.

Feature-Gradient Energy Visualization: Fig. 5 qualitatively supports these findings. Supervised classification produces diffuse, texture-dominated responses with weak boundary correspondence even after full fine-tuning. MAE and DINO

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TABLE II

SOTA COMPARISON ON ISPRS VAIHINGEN AND POTSDAM (%). EACH CELL REPORTS VAIHINGEN/POTSDAM. BEST AND SECOND-BEST RESULTS ARE HIGHLIGHTED IN BOLD AND UNDERLINE, RESPECTIVELY

Structures	Method	Trainable Params (M)	Per-class F1 (%)					Overall (%)		
			Imp. Surf.	Building	Low Veg.	Tree	Car	mF1	mIoU	OA
CNN-based	FCN8s [57]	30.03	95.48/90.95	92.12/92.84	82.03/85.33	88.80/86.25	79.96/92.80	87.68/89.63	78.55/81.37	89.22/88.34
	DeepLabv3+ [21]	59.30	92.27/93.15	95.40/95.43	83.56/86.45	89.22/87.78	85.36/95.21	89.16/91.60	80.72/84.73	90.27/90.07
	DANet [58]	66.40	91.38/91.94	94.10/96.35	83.09/86.00	89.02/87.67	76.79/86.01	86.88/89.59	77.32/81.40	89.20/89.73
	PSPNet [59]	46.60	91.17/90.79	94.64/95.15	82.92/86.57	88.99/87.32	73.41/91.05	86.23/90.18	76.52/82.25	89.49/89.31
	BiSeNet [60]	49.00	89.10/91.07	91.30/95.12	80.91/85.81	86.91/86.63	73.07/91.37	84.26/90.00	73.34/81.99	87.10/89.06
Transformer-based	UNetFormer [23]	11.68	96.80/93.78	95.49/96.20	83.61/87.52	89.36/88.44	87.62/96.17	90.58/92.42	83.15/86.13	90.99/91.21
	BANet [61]	12.73	92.20/93.30	95.20/96.70	83.80/87.40	89.90/89.10	86.80/96.00	89.58/92.50	81.36/86.26	90.50/91.00
	MA-Net [24]	35.68	93.02/93.24	95.47/95.64	84.64/86.83	89.98/88.15	88.95/95.83	90.41/91.94	82.71/85.30	90.96/90.82
	A^2-FPN [25]	22.82	96.59/93.43	95.30/96.12	83.27/86.89	89.37/88.38	86.93/95.17	90.29/92.00	82.69/85.40	90.78/90.83
	CMTFNet [26]	30.07	96.74/93.28	95.70/97.07	84.23/87.08	89.76/88.42	88.31/95.93	90.95/92.36	83.74/86.05	91.15/90.92
	MCSNet [62]	26.00	93.52/93.59	95.85/97.09	85.83/88.15	**91.12**/89.63	90.48/95.98	91.36/92.89	84.27/86.92	91.97/91.40
	SparseFormer [28]	42.52	96.72/93.56	95.65/96.00	84.42/87.55	89.93/89.20	88.85/96.15	91.11/92.49	84.00/86.23	91.21/91.56
	SCSM [29]	30.19	94.48/93.87	95.84/97.16	84.69/90.44	90.25/88.74	92.58/96.76	91.57/93.39	84.68/87.79	92.22/92.13
	CenterSeg [27]	30.60	94.80/94.24	96.20/**97.68**	85.34/**90.95**	90.78/89.03	**92.76**/96.90	91.98/93.76	85.37/88.44	92.56/92.54
	MPKG-Net [63]	171.16	93.69/95.42	**97.11**/97.58	85.91/89.70	90.02/89.80	89.77/97.47	91.30/93.99	84.22/88.88	92.34/93.94
SAM-based	SAM-Assisted [34]	96.14	93.68/95.19	96.88/96.89	84.57/88.50	90.20/88.39	89.64/96.99	90.99/93.19	83.74/87.50	91.80/93.00
	SAM-LST [64]	11.76	96.20/91.80	94.20/93.80	83.90/85.70	89.30/86.90	86.20/94.80	89.96/90.60	82.08/83.02	92.40/89.00
	MSA-SAM [35]	24.40	**97.23**/94.79	96.16/97.28	85.84/88.33	90.53/89.44	90.41/96.59	92.03/93.29	85.52/87.64	93.90/92.10
Freq-Enhanced	DualM [43]	33.90	93.20/93.20	95.50/97.00	84.60/86.90	89.80/88.80	88.90/95.90	90.40/92.36	82.69/86.05	91.40/90.80
	SFFNet [44]	34.16	96.86/93.40	95.57/96.62	84.79/87.73	90.13/89.20	89.27/96.33	91.32/92.66	84.34/86.53	91.36/91.72
	XNet [65]	41.42	96.81/93.25	95.63/96.57	84.96/87.83	90.32/89.43	89.29/96.43	91.40/92.70	84.46/86.60	91.52/91.83
	AFENet [45]	20.23	96.90/94.57	95.72/96.86	85.07/88.17	90.64/89.78	89.37/96.83	91.54/93.24	84.69/87.55	91.67/92.03
	SynFSNet [46]	68.43	97.20/93.92	96.40/97.35	85.31/88.58	90.58/89.64	91.27/96.95	92.15/93.29	85.74/87.64	93.85/91.87
Ours	**CFP-SAM**	**6.39**	95.74/**95.79**	97.00/97.40	**86.07**/89.84	91.10/**90.30**	91.48/**97.74**	**92.28/94.21**	**85.90/89.26**	**94.18/93.97**

recover partial spatial structure but still show diffuse or locally edge-biased activations. SAM exhibits clearer object contours before adaptation and largely preserves them under rank-1 LoRA, consistent with its small fine-tuning gap.

For SAMRS, qualitative and quantitative behaviors are not fully aligned. Rank-1 LoRA produces weak responses and poor quantitative results. Full fine-tuning restores more visible boundary structures but remains substantially below its supervised-classification counterpart, suggesting that boundary clarity alone is insufficient without semantic discriminability and encoder–decoder compatibility.

C. Comparison With State-of-the-Art Methods (RQ2)

To provide a comprehensive comparison, we include representative CNN-based, Transformer-based, SAM-based, and frequency-enhanced methods on remote sensing semantic segmentation benchmarks.

Overall Quantitative Comparison: Tables II and III report the comparison results on the ISPRS and urban-village benchmarks. CFP-SAM achieves the best overall performance across all four datasets on the primary metrics. In particular, it obtains mIoU scores of 85.90% and 89.26% on Vaihingen and Potsdam, respectively, and foreground IoU scores of 74.46% and 81.48% on Beijing and Xi'an, respectively. CFP-SAM also uses only 6.39 M trainable parameters, the smallest among all compared methods. These results demonstrate that CFP-SAM improves segmentation accuracy without relying on large-scale parameter updates, achieving a favorable balance between segmentation performance and adaptation efficiency.

TABLE III

PERFORMANCE COMPARISON ON BEIJING/XI'AN URBAN-VILLAGE DATASETS (%). EACH CELL REPORTS BEIJING/XI'AN. BOLD: BEST. UNDERLINE: SECOND-BEST

Method	Trainable Params (M)	Metrics (%)			
		IoU	F1	Recall	Precision
FCN8s [57]	30.03	66.04/72.03	79.55/83.74	77.63/81.85	81.56/85.72
PSPNet [59]	46.60	69.67/76.03	82.12/86.38	80.21/84.48	84.13/88.37
DeepLabV3+ [21]	59.30	65.06/66.91	78.83/80.18	76.91/78.26	80.85/82.19
SegFormer [22]	13.70	69.97/71.34	82.33/83.27	80.44/81.39	84.32/85.24
SegNeXt [66]	13.90	70.74/75.96	82.86/86.34	80.95/84.41	84.87/88.35
UVLens [67]	44.41	62.40/68.72	76.85/81.46	74.96/79.53	78.84/83.48
RSPrompter [33]	14.34	46.23/56.84	63.23/72.48	61.37/70.62	65.21/74.45
SAM2-UNet [68]	11.40	53.98/60.21	70.11/75.16	68.29/73.27	72.03/77.16
UV-SAM [15]	14.45	69.73/71.97	82.17/83.70	80.25/81.78	84.18/85.72
UV-Mamba [69]	7.70	72.79/78.02	84.25/87.64	82.34/85.72	**86.26**/89.68
CFP-SAM (Ours)	**6.39**	**74.46/81.48**	**85.36/89.80**	**86.06/88.56**	84.67/**91.07**

Comparison With Transformer-Based Methods: Compared with representative Transformer-based methods, CFP-SAM achieves stronger overall segmentation performance with substantially fewer trainable parameters. Recent methods such as SCSM [29], CenterSeg [27], and MPKG-Net [63] improve remote sensing segmentation by enhancing spatial context modeling, category representation, or cross-patch semantic modeling. On the contrary, CFP-SAM regularizes adaptation by suppressing low-frequency style dependence during fine-tuning. It surpasses MPKG-Net, the strongest Transformer-based baseline in mIoU, by 1.68 and 0.38 percentage points on Vaihingen and Potsdam, respectively, while using only approximately 3.7% of its trainable parameter count. These results suggest that controlling

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style-biased adaptation provides a parameter-efficient alternative to adding complex task-specific context-modeling modules.

Comparison With SAM-Based Methods: SAM-based baselines adapt SAM priors mainly by adding auxiliary branches [64], multiscale adapters [35], or SAM-derived supervision cues [34]. On the contrary, CFP-SAM keeps the SAM-pretrained encoder frozen and regularizes fine-tuning via low-frequency perturbations and semantic-consistency constraints. Compared with MSA-SAM, the strongest baseline in this group, CFP-SAM improves mIoU by 0.38 and 1.62 percentage points on Vaihingen and Potsdam, respectively, while using approximately one-quarter of its trainable parameters. These results indicate that suppressing low-frequency style dependence is critical for parameter-efficient SAM adaptation in remote sensing segmentation.

Comparison With Frequency-Enhanced Methods: Frequency-enhanced methods typically embed frequency-domain operations into the forward feature-extraction process. For example, SynFSNet [46] uses AFSIFormer blocks to introduce adaptive frequency–spatial interaction attention into the forward feature-extraction path, while AFENet [45] adaptively separates and modulates high- and low-frequency components for feature enhancement. CFP-SAM uses frequency-domain perturbation as a training-time regularization signal. It perturbs low-frequency feature amplitudes and enforces semantic consistency under perturbed features, so LLFP is disabled during inference and introduces no LLFP-related inference overhead. CFP-SAM slightly outperforms SynFSNet on Vaihingen and shows a larger advantage on Potsdam while requiring less than one-tenth of its trainable parameters. These results show that suppressing low-frequency bias during training is an effective, parameter-efficient way to exploit frequency-domain information.

Results on Urban-Village Datasets: Table III further validates CFP-SAM on highly imbalanced binary segmentation. Urban-village identification in Beijing and Xi'an involves sparse foregrounds, irregular shapes, and ambiguous boundaries, imposing high demands on localization and recall. CFP-SAM achieves 74.46% and 81.48% IoU on Beijing and Xi'an, respectively, surpassing the strongest baseline, UV-Mamba [69], by 1.67 and 3.46 percentage points. It also obtains the best $F1$-score and recall on both datasets, showing that reducing reliance on low-frequency appearance and regional style cues improves recognition stability in sparse and boundary-ambiguous scenes. RSPrompter [33] shows relatively low IoU in this setting. This is because RSPrompter is designed around instance-oriented anchor-based and query-based prompts, which are better suited to discrete, well-bounded objects than to region-level settlement patterns with weak and irregular boundaries.

Qualitative Analysis: Figs. 6 and 7 show the comparison of the segmentation results on the ISPRS datasets. On Vaihingen, competing methods produce fragmented predictions around small objects and category-ambiguous regions, especially near tree–low-vegetation transitions and dense vehicle areas. CFP-SAM generates more coherent regions and better preserves local boundaries, which is consistent with its leading mIoU and mF1 scores. On Potsdam, CFP-SAM maintains clearer boundaries in complex layouts that include impervious surfaces, vegetation, buildings, and cars. This observation is consistent with its leading $F1$ scores for the tree and car classes and its best overall mIoU on the Potsdam dataset. Fig. 8 shows qualitative comparisons on the Beijing and Xi'an urban-village datasets. SegFormer, UV-SAM, and UV-Mamba tend to miss irregular foreground regions or produce incomplete boundaries, whereas CFP-SAM recovers more complete urban-village regions with smoother contours. These qualitative results are consistent with the higher recall and IoU achieved by CFP-SAM on both datasets.

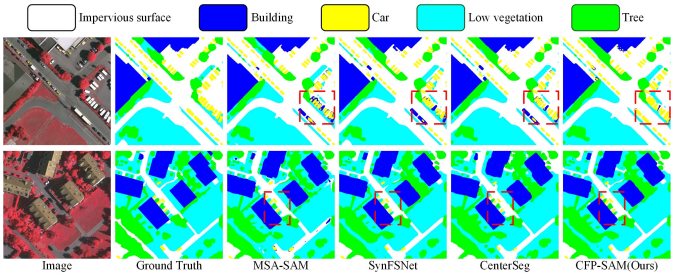


Fig. 6. Qualitative comparison with strong SOTA methods on ISPRS Vaihingen. (Left to right) Input image, ground truth, MSA-SAM, SynFSNet, CenterSeg, and CFP-SAM (ours).

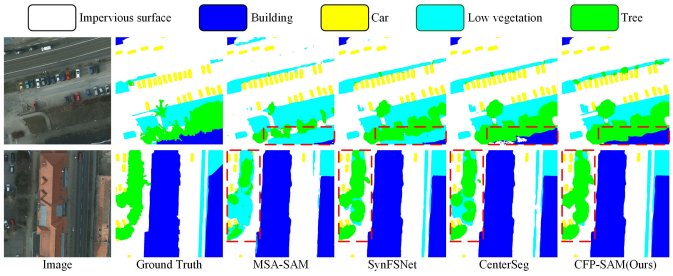


Fig. 7. Qualitative comparison with strong SOTA methods on ISPRS Potsdam. (Left to right) Input image, ground truth, MSA-SAM, SynFSNet, CenterSeg, and CFP-SAM (ours).

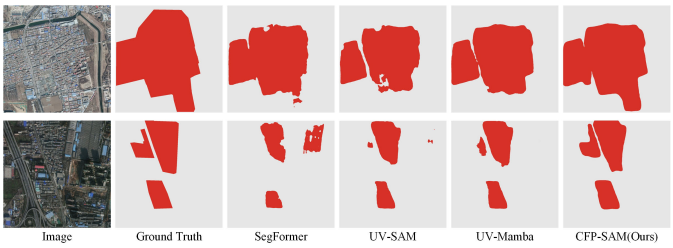


Fig. 8. Qualitative comparison on Beijing (row 1) and Xi'an (row 2) urban-village datasets. (Left to right) Input, ground truth, SegFormer, UV-SAM, UV-Mamba, and CFP-SAM (ours).

D. Ablation Study (RQ3)

To identify the sources of CFP-SAM's performance gains, we conduct ablation studies on the ISPRS Potsdam dataset, with selected results reported on the Xi'an dataset. All variants use the same SAM-pretrained Hiera-Large encoder with

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TABLE IV
ABLATION STUDY OF CORE COMPONENTS ON ISPRS POTSDAM (%)

Variant	Components					Potsdam (%)		
	MS-Head	Rand-LF	LLFP + $\mathcal{D}_{\text{style}}$	$\mathcal{L}_{\text{seg}}^{\text{pert}}$	$\mathcal{L}_{\text{cons}}$	mIoU	mF1	OA
Baseline (f_4 only)						88.05	93.53	93.50
+MS-Head (f_1–f_4)	✓					88.55	93.81	93.64
+Rand-LF	✓	✓		✓		88.73	93.91	93.68
+LLFP + $\mathcal{D}_{\text{style}}$	✓		✓	✓		89.02	94.07	93.80
w/o pert. sup.	✓		✓		✓	89.19	94.17	93.90
KL cons.	✓		✓	✓	✓	88.90	94.01	93.73
w/o stop-grad	✓		✓	✓	✓	89.05	94.09	93.83
Full CFP-SAM	✓		✓	✓	✓	**89.26**	**94.21**	**93.97**

TABLE V
EFFECT OF LLFP INJECTION STAGES (%)

Injection Stages	Potsdam			Xi'an			
	mIoU	mF1	OA	IoU	F1	Recall	Precision
f_4 only	88.62	93.84	93.53	80.64	89.28	89.85	88.72
f_3–f_4	88.85	93.97	93.73	80.70	89.32	87.68	91.02
f_2–f_4	89.11	94.13	93.82	81.21	89.63	91.10	88.20
f_1–f_4 (Full)	**89.26**	**94.21**	**93.97**	**81.48**	**89.80**	**88.56**	**91.07**

TABLE VI
ABLATION STUDY OF LLFP INTERNAL DESIGN ON ISPRS POTSDAM (%)

Variant	Design		Potsdam (%)		
	Learnable r_c	Soft Mask	mIoU	mF1	OA
Frequency separation design					
Fixed r_c + Hard mask			88.76	93.92	93.66
Learnable r_c + Hard mask	✓		89.12	94.13	93.86
Fixed r_c + Soft mask		✓	88.89	93.99	93.73
Learnable r_c + Soft mask	✓	✓	89.26	94.21	93.97
Perturbation granularity					
Shared γ, β (scalar)	✓	✓	89.07	94.10	93.83
Per-channel γ, β (Full)	✓	✓	**89.26**	**94.21**	**93.97**

TABLE VII
EFFECT OF PERTURBATION FREQUENCY RANGE ON ISPRS POTSDAM (%)

Variant	Range	mIoU	mF1	OA
HF-only perturbation	(0.55, 1.00]	88.31	93.66	93.39
All-frequency perturbation	(0.00, 1.00]	88.91	94.01	93.67
Full CFP-SAM	(0.05, 0.45]	**89.26**	**94.21**	**93.97**

LoRA adaptation; the ablations vary the segmentation head, perturbation strategy, loss terms, or LLFP design choices.

Analysis of Core Component Effectiveness: Table IV shows the progressive contribution of each component. Starting from the single-scale baseline at 88.05% mIoU, the multiscale segmentation head improves to 88.55%, indicating that hierarchical feature fusion better preserves spatial information across semantic levels. Adding Rand-LF further improves mIoU to 88.73%, confirming that low-frequency perturbation itself benefits robustness. Replacing Rand-LF with LLFP and style-discrepancy maximization raises mIoU to 89.02%, showing that learnable perturbations explore more effective low-frequency style variations. Removing perturbed-branch supervision while retaining semantic consistency yields 89.19% mIoU, whereas Full CFP-SAM reaches 89.26%, showing that perturbed-branch supervision further improves performance alongside consistency regularization.

We further evaluate two design choices in the consistency loss. Replacing squared ℓ_2 consistency with KL divergence reduces mIoU to 88.90%, suggesting that KL divergence may place stronger penalties when the perturbed branch assigns very low probabilities to classes retained by the online clean prediction. Removing stop-gradient from the clean branch reduces mIoU to 89.05%, supporting its role in maintaining the clean prediction as a stable anchor and preserving directional regularization on the perturbed branch.

Analysis of LLFP Injection Stages: Table V examines the effect of applying LLFP across encoder stages. Injecting LLFP only at f_4 yields the lowest performance. Extending the injection range from f_3–f_4 to f_2–f_4 improves mIoU on Potsdam from 88.85% to 89.11% and IoU on Xi'an from 80.70% to 81.21%. Applying LLFP to all stages f_1–f_4 achieves the best primary metrics. Although f_2–f_4 gives higher recall on Xi'an, the full-stage design achieves better IoU, $F1$-score, and Precision, indicating a better balance between foreground completeness and false-positive control.

Analysis of LLFP Internal Design: Table VI evaluates frequency separation and perturbation granularity. For frequency separation, a learnable per-channel radius improves mIoU from 88.76% to 89.12% compared with a fixed radius, indicating that adaptive boundaries better match channel-specific style distributions. Adding the raised-cosine soft mask further improves mIoU to 89.26%, suggesting that its smooth transition alleviates abrupt frequency truncation and stabilizes perturbation. For perturbation granularity, per-channel modulation at 89.26% outperforms shared scalar modulation at 89.07%, showing that channelwise modulation better captures heterogeneous frequency statistics.

Analysis of Perturbation Frequency Range: Table VII illustrates the comparison of the perturbation frequency ranges. High-frequency-only perturbation yields the lowest mIoU of 88.31%, indicating that disturbing boundary- and texture-related components degrades segmentation performance. All-frequency perturbation improves mIoU to 88.91% but still underperforms the low-frequency-only design, which achieves 89.26% mIoU. These results confirm that the performance gain comes from targeted regularization of low-frequency style components rather than arbitrary perturbation across the spectrum.

Visualization of LLFP Mechanism: Fig. 9 visualizes the frequency-selective behavior of LLFP. The learned soft mask $\mathbf{M}$ concentrates around the spectral center, consistent with the radius constraint $[r_{\min}, r_{\max}] = [0.05, 0.45]$. The amplitude difference maps $|\tilde{\mathbf{A}} - \mathbf{A}|$ show that perturbation energy is concentrated at low frequencies and attenuates toward high-frequency regions, while the perturbed feature maps retain the major spatial structures.

Analysis of LoRA Rank Sensitivity: Fig. 10 shows that performance generally improves up to rank 32 across the four datasets and then saturates or slightly declines. We therefore

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TABLE VIII

SOURCE-ONLY GENERALIZATION BETWEEN ISPRS POTSDAM RGB AND VAIHINGEN IRRG (%). $P(r) \rightarrow V$ AND $V \rightarrow P(r)$ DENOTE THE TWO TRANSFER DIRECTIONS. BOLD AND UNDERLINED VALUES INDICATE THE BEST AND SECOND-BEST RESULTS, RESPECTIVELY

Method	Backbone	$\mathbf{P(r) \rightarrow V}$						$\mathbf{V \rightarrow P(r)}$					
		Imp.S.	Bldg.	L.Veg.	Tree	Car	mIoU	Imp.S.	Bldg.	L.Veg.	Tree	Car	mIoU
DAFormer [70]	MiT-B5	64.2	73.4	4.5	9.7	42.2	38.80	46.0	59.4	12.6	5.8	63.5	37.46
HRDA [71]	MiT-B5	67.1	66.9	4.1	17.5	43.0	39.72	54.7	54.7	11.6	14.4	72.3	41.54
S12-MAE [72]	ViT-S	9.0	25.5	1.3	0.0	5.7	8.30	40.9	29.9	9.1	2.0	21.1	20.60
SatMAE [37]	ViT-L	53.6	26.4	3.5	0.6	20.1	20.84	45.0	38.0	20.9	4.2	29.4	27.50
ScaleMAE [38]	ViT-L	43.3	30.9	1.1	1.8	7.4	16.90	43.9	51.9	25.2	0.8	35.2	31.40
MTP [39]	ViT-L	51.0	64.8	4.4	7.5	56.8	36.90	58.5	76.5	44.4	10.8	82.0	54.44
Rein [73]	ViT-L	79.1	91.2	37.0	62.2	61.1	66.12	70.4	77.4	58.6	13.9	78.5	59.76
CrossEarth [53]	ViT-L	78.1	89.3	55.2	72.5	**61.9**	71.40	**72.5**	73.6	**58.7**	22.4	81.1	61.66
Baseline	Hiera-L	76.74	89.82	50.68	74.48	46.93	67.73	68.09	81.69	45.17	21.13	83.53	59.92
CFP-SAM (Ours)	Hiera-L	**81.36**	**91.75**	**58.16**	**75.42**	52.62	**71.86**	70.27	**84.23**	46.32	**32.87**	**83.92**	**63.52**

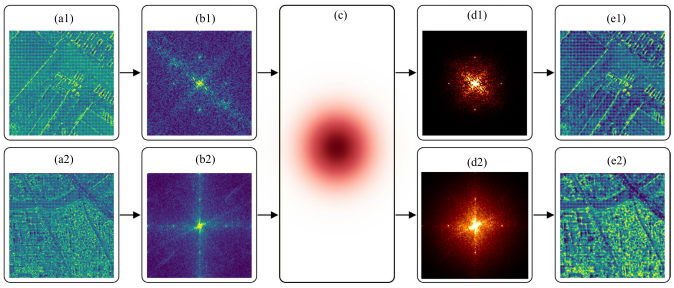


Fig. 9. Visualization of LLFP on encoder feature samples. Columns show the original feature map, FFT amplitude spectrum, learned low-frequency soft mask $\mathbf{M}$, amplitude difference $|\tilde{\mathbf{A}} - \mathbf{A}|$, and perturbed feature map. (a1) and (a2) Feature map (before). (b1) and (b2) FFT amplitude spectrum. (c) Soft mask M. (d1) and (d2) Amplitude difference $|\tilde{\mathbf{A}} - \mathbf{A}|$. (e1) and (e2) Feature map (after).

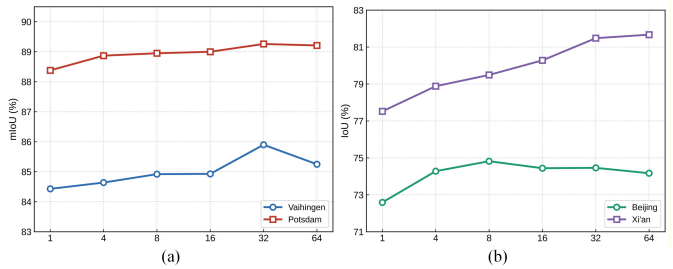


Fig. 10. Effect of LoRA rank on segmentation performance. (a) IoU on Beijing and Xi'an urban-village datasets. (b) mIoU on ISPRS Vaihingen and Potsdam datasets. Rank 32 provides the best overall tradeoff across datasets.

adopt rank 32 as the default tradeoff between adaptation capacity and parameter efficiency.

E. Source-Only Cross-Domain Generalization (RQ4)

To evaluate cross-domain transferability, we conduct source-only generalization experiments under two protocols. The first evaluates bidirectional transfer between ISPRS Potsdam (RGB) and Vaihingen (IRRG), where the shift spans both geographic distribution and channel semantics, with the first channel changing from red to near-infrared. The second evaluates cross-city transfer between the Beijing and Xi'an urban-village datasets, where the channel configuration is consistent, and the shift is primarily geographic and appearance-related. In all settings, models are trained only with labeled source-domain data, and target-domain images are used only for testing. The baseline shares the same Hiera-L backbone, LoRA configuration, and multiscale FPN segmentation head as CFP-SAM, differing only in the removal of LLFP and the associated dual-constraint training strategy; the performance gaps between the two mainly reflect the contribution of the proposed frequency-aware training approach.

ISPRS Cross-Domain Generalization: Table VIII reports results on $P(r) \rightarrow V$ and $V \rightarrow P(r)$. Most compared methods suffer substantial degradation under this source-only protocol. MTP [39], despite remote sensing pretraining, obtains only 36.90% mIoU on $P(r) \rightarrow V$, indicating that remote sensing pretraining alone is insufficient to bridge severe cross-domain gaps. Baseline already achieves 67.73% and 59.92% mIoU in the two directions, demonstrating strong structural transferability of SAM pretraining. CFP-SAM further improves to 71.86% and 63.52% mIoU, with gains of 4.13 and 3.60 percentage points, respectively.

CFP-SAM also compares favorably with CrossEarth [53], a method specifically designed for cross-domain remote sensing segmentation, achieving 71.86% versus 71.40% on $P(r) \rightarrow V$ and 63.52% versus 61.66% on $V \rightarrow P(r)$. Gains are stronger on structurally stable categories such as buildings and trees; Car IoU on $P(r) \rightarrow V$ and Low Veg. IoU on $V \rightarrow P(r)$ remain more limited, suggesting that class-specific spectral discrepancies pose additional challenges under cross-band transfer.

Feature-Space Analysis: Fig. 11 visualizes decoder features extracted from target-domain images. CFP-SAM produces more compact, better-separated class clusters in both directions of transfer. On $P(r) \rightarrow V$, the silhouette coefficient increases from 0.305 to 0.346, and the Davies–Bouldin index decreases from 1.17 to 0.98; on $V \rightarrow P(r)$, the two metrics change from 0.301 to 0.346 and from 1.26 to 1.01, respectively. These results indicate that CFP-SAM strengthens semantic discriminability in the transferred feature space, consistent with the class-level improvements in Table VIII.

Urban-Village Cross-City Generalization: Table IX reports cross-city results on $B \rightarrow X$ and $X \rightarrow B$. General and

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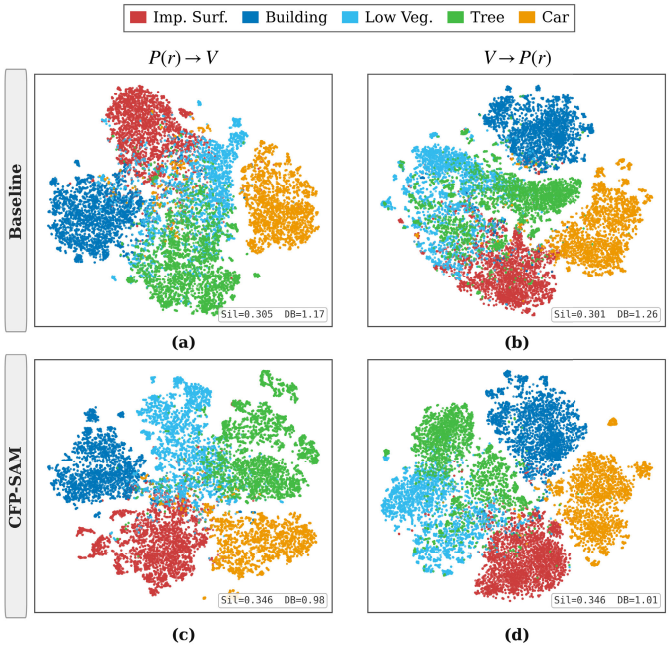


Fig. 11. t-SNE visualization of target-domain decoder features. (a) and (b) SAM Hiera-L baseline. (c) and (d) CFP-SAM. Sil and DB denote the silhouette coefficient and the Davies–Bouldin index, respectively.

TABLE IX

SOURCE-ONLY CROSS-CITY GENERALIZATION BETWEEN THE BEIJING AND XI'AN URBAN-VILLAGE DATASETS (%). $B \rightarrow X$ AND $X \rightarrow B$ DENOTE THE TWO TRANSFER DIRECTIONS. BOLD AND UNDERLINED VALUES INDICATE THE BEST AND SECOND-BEST RESULTS, RESPECTIVELY

Method	Backbone	B→X		X→B	
		IoU	F1	IoU	F1
PSPNet [59]	ResNet-50	63.69	77.82	47.12	64.06
DeepLabV3+ [21]	ResNet-101	62.46	76.89	45.25	62.30
SegFormer [22]	MiT-B1	65.73	79.32	49.93	66.61
SegNeXt [66]	MSCAN-T	64.30	78.27	47.59	64.49
UV-SAM [15]	ViT-L	68.33	81.19	52.18	68.58
UV-Mamba [69]	SSM-based	70.48	82.68	56.32	72.06
Baseline	Hiera-L	<u>76.44</u>	<u>86.65</u>	<u>68.65</u>	<u>81.41</u>
CFP-SAM (Ours)	Hiera-L	**80.46**	**89.17**	**73.33**	**84.62**

urban-village-specific methods, including UV-SAM [15] and UV-Mamba [69], show limited transferability. Baseline substantially outperforms them, reaching 76.44% and 68.65% IoU in the two directions. CFP-SAM further improves to 80.46% and 73.33% IoU, yielding gains of 4.02 and 4.68 percentage points, showing that the proposed low-frequency perturbation strategy also benefits morphologically irregular binary segmentation with sparse foregrounds and ambiguous boundaries.

V. CONCLUSION

This article studied the transferability of pretrained models for remote sensing semantic segmentation from a frequency-domain perspective. We identified low-frequency bias as a factor that weakens the transfer of high-frequency structural cues and observed that SAM's prompt-driven pretraining preserves stronger high-frequency responses compared with the other evaluated pretraining paradigms. Based on this observation, we proposed CFP-SAM, which uses LLFPs, style-discrepancy maximization, and semantic-consistency constraints to suppress style-dependent adaptation. Experiments across four benchmarks showed that CFP-SAM consistently improves source-domain segmentation accuracy and source-only cross-domain generalization while using limited trainable parameters and incurring no additional inference overhead. Future work will explore class-adaptive perturbation and extensions to multisource and open-vocabulary remote sensing segmentation.

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